Bioactives

PEPHA®-PROTECT

Powerful bodyguard for naturally protected skin

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NUTRITION · HEALTH · SUSTAINABLE LIVING

DSM
BRIGHT SCIENCE. BRIGHTER LIVING.

A bodyguard from the desert

Light is the source of life. But while infrared and visible light normally provide us with warmth and well-being, ultraviolet light can be very harmful to the skin. Aging or stressed skin is especially vulnerable because the regenerative power of its cells is reduced.

The plants of the African desert have learned how to protect themselves against the scorching sun. Research at DSM on the watermelon (*Citrullus lanatus*), has shown just how it shields itself from extreme heat, drought, and intensive UV radiation: its fruit produces tough defensive compounds, so-called compatible solutes, to protect important cellular structures.

PEPHA®-PROTECT, a highly purified *Citrullus lanatus* extract from a species whose origins lie in Kalahari desert, harnesses the action of the watermelon to strengthen the natural defenses of human skin. It helps prevent DNA damage in skin cells in two ways: by preventing direct damage from UV light and by inhibiting the release of free radicals induced by UV light. PEPHA®-PROTECT is thus a powerful bodyguard providing tough protection for the skin.

Key facts

UNIQUE PRODUCT FEATURES

- Unique highly purified *Citrullus lanatus* extract with high concentration of compatible solutes
- Significant DNA protection *in vitro* against UV damage and damage induced by free radicals
- Significant protection of DNA nucleotides *in vitro* from oxidation processes
- *In vivo* proven 25% increase in skin protection

BENEFITS

- Strengthens skin's own cell regeneration ability and helps to protect from DNA damage
- Protects the skin from UV induced inflammation and irritation, providing a more even skin texture
- Helps prevent premature skin aging, the skin's appearance remains younger

COSMETIC APPLICATION

- Anti-aging day care claiming DNA protection
- Facial care with regenerating and repairing properties
- Skin protection products in combination with UVA/UVB filters

SUGGESTED CONCENTRATION

1-3% PEPHA®-PROTECT in cosmetic formulations

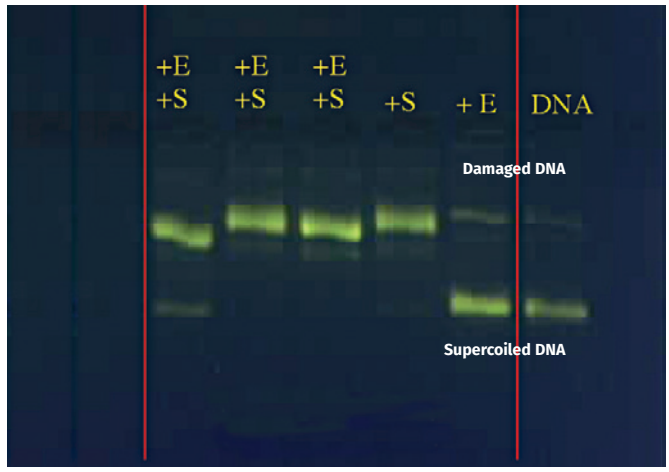
INCI NAME

Citrullus lanatus (Watermelon) Fruit Extract, Citrulline

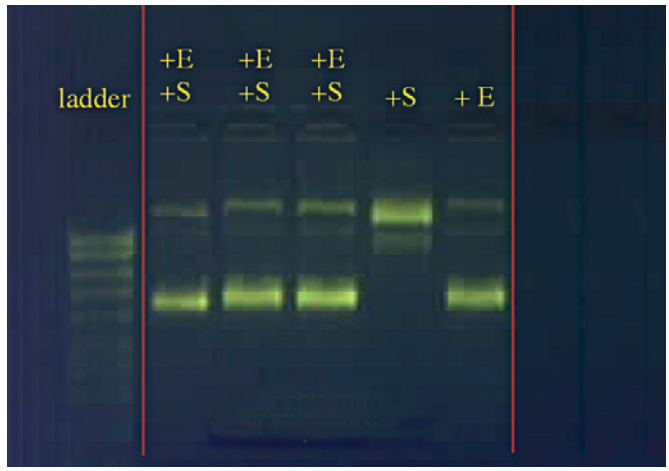
Technology

The active compounds of the melon's defense system, the compatible solutes, are water soluble molecules. A standard hydrophilic watermelon extract as well as extracts purified by several purification steps were therefore produced. A plasmid assay combined with electrophoresis was applied to confirm the activity of these extracts.

The assay was performed on isolated plasmid DNA, which was stressed by peroxide to induce DNA damage. Damaged DNA changed its conformation and could be distinguished from undamaged DNA (supercoiled DNA) by electrophoresis: the undamaged and supercoiled DNA is more compact and migrates faster from the top downwards compared to damaged DNA. Supercoiled (undamaged) DNA and damaged DNA become therefore visible as separate bands in the gel.



Standard watermelon extract



Highly pure watermelon extract – PEPHA®-PROTECT

+ E: plus melon extract. **+ S:** plus H2O2 stress

Results clearly showed that only the highly purified watermelon extract, used to formulate PEPHA®-PROTECT, can provide almost complete DNA protection *in vitro*.

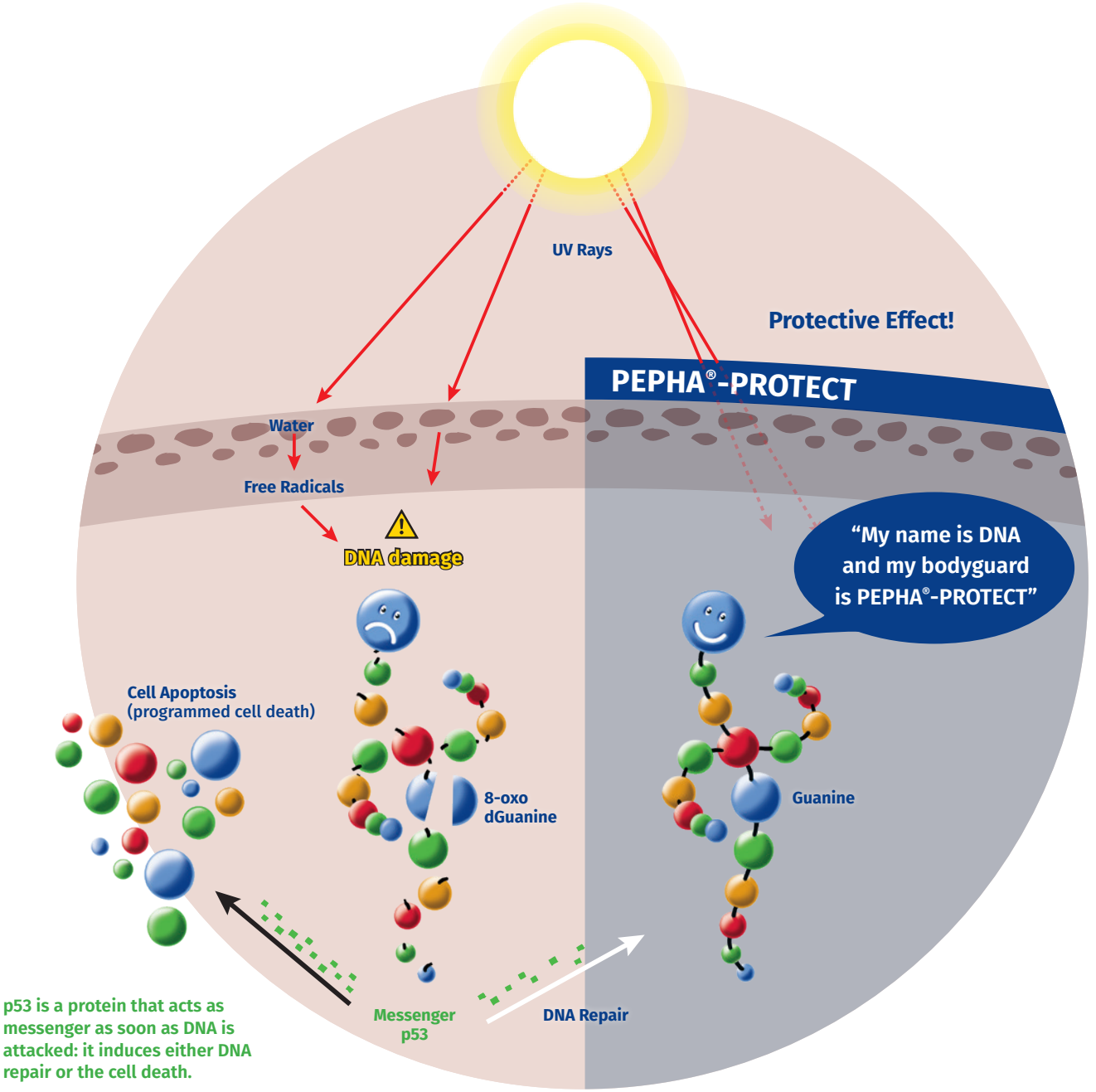
Several *in vitro* tests and an *in vivo* study have demonstrated the outstanding capabilities of PEPHA®-PROTECT to protect the skin.

Mechanism

The DNA (DeoxyriboNucleic Acid) molecule encodes the genetic instructions needed for life to develop and function properly. It forms a double-stranded helix structure, with each strand consisting of two long polymers made from repeating units called nucleotides (guanine, adenine, thymine, and cytosine).

Although normally very stable, in skin the DNA can be damaged by UVA (ultra violet A) radiation from the sun. This damage results either directly from the energy of the radiation, or indirectly from the release of free radicals, which can cleave DNA strands in the skin.

Damaged DNA leads to erroneous genetic function and ultimately to accelerated skin aging, characterized by wrinkles, loss of elasticity and sagging, or pigmentation disorders.



PEPHA®-PROTECT is a highly purified watermelon (*Citrullus lanatus*) extract containing high quantities of precious compatible solutes — naturally occurring vitamins, carbohydrates, and specific amino acids, such as citrulline. These compounds act together to protect important cellular structures, including DNA.

UVA modification of DNA in skin cells triggers different responses which include activation of a dedicated protein, p53, by a process known as phosphorylation. This initiates either DNA repair or apoptosis (programmed cell death). Another effect of UV irradiation is oxidation of the guanine nucleotide to form 8-oxo guanine, which can lead to the breakdown of DNA strands.

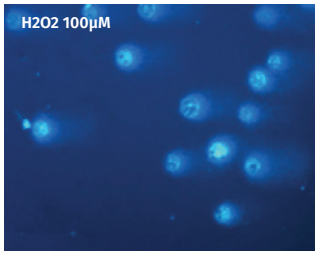
Several advanced in vitro assays based on the quantification of both 8-oxo guanine and phospho-p53 protein demonstrate the tremendous capacity of PEPHA®-PROTECT to protect skin DNA from irreversible damage, keeping cells in good shape and thereby sustaining skin's natural beauty.

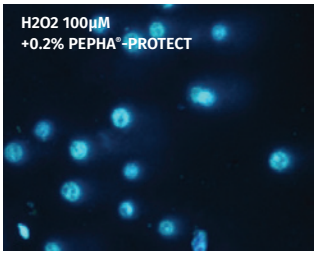
Efficacy *in vitro*

Comet assay

Carried out on fibroblasts and on a human skin model consisting of keratinocytes, in order to show DNA protecting capacity of PEPHA®-PROTECT.

The keratinocytes have been stressed by hydroperoxide, which leads to DNA breakdowns. During electrophoresis, small sections of damaged DNA may migrate out of the nucleus into the cytoplasm and thereby cause the so-called comet effect. A longer tail indicates more DNA damage.

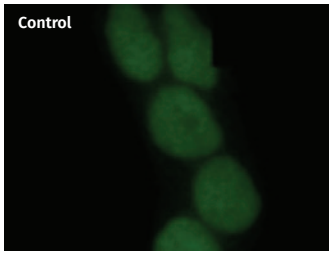


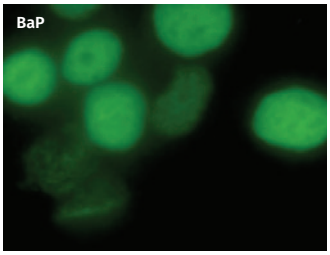


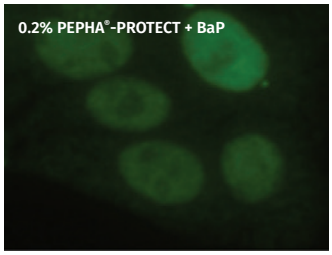
P53 assay

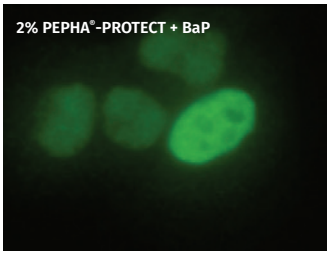
Healthy cells naturally contain small amounts of protein p53 (phosphorylated protein p53). As soon as cellular DNA damage is detected, the protein p53 is activated by phosphorylation. Depending on the severity of the DNA damage, phospho-p53 initiates DNA repair or apoptosis. The amount of phospho-p53 is therefore correlated to the extent of DNA damage.

Keratinocytes were stressed with benzoapyrene (BaP) and the resulting phospho-p53 was quantified by fluorescence.

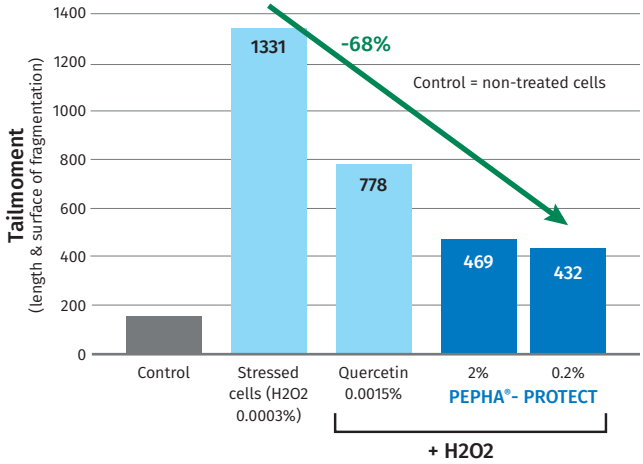






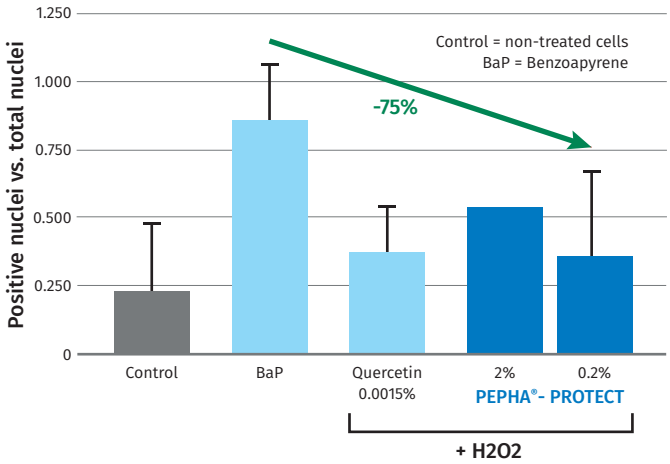


Comet assay in human skin model



Results: 69% DNA protection with 0.2% PEPHA®-PROTECT on a human skin model.

Phospho-p53 staining in keratinocytes

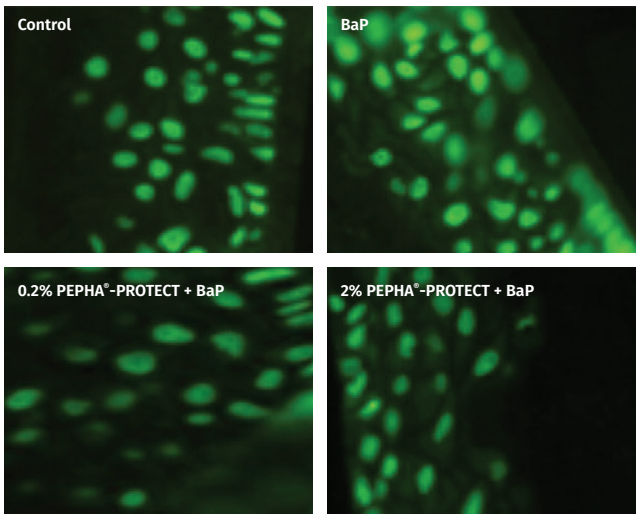


Results: 75% decrease in phospho-p53 production with 0.2% PEPHA®-PROTECT.

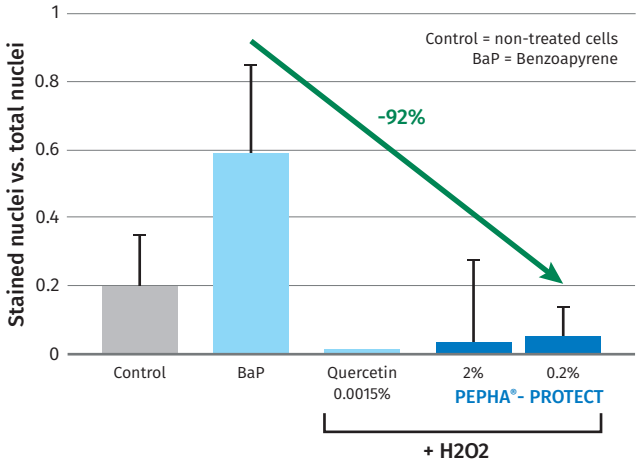
Efficacy *in vivo*

8-oxo-dG assay

DNA consists of a double helix, each strand of which is composed of nucleotide (guanine, adenine, thymine, and cytosine) sequences. After UV irradiation the guanine base is oxidized at the C8 position and is transformed into 8-oxo-guanine, which can be detected by fluorescence with appropriate antibodies. Such oxidation leads to DNA strand breaks.



8-oxo-DG staining on reconstituted epidermis



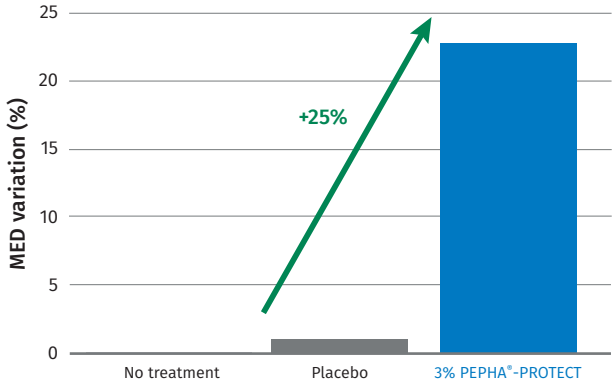
Results: 92% decrease of 8-oxo-dG production with 0.2% PEPHA®-PROTECT.

Reinforcing the skin's defense system

In a 28-day study with PEPHA®-PROTECT, Minimal Erythema Dose (MED) was measured. This is the UV radiation level needed to cause skin reddening. The higher the MED for a substance, the greater its protective effect.

In this study a cream containing 3% PEPHA®-PROTECT was applied twice daily. Measured parameter: minimum dosage of UV light needed to turn the skin red at a read-out point 24 h later.

Improvement of the protective effect on UV irradiations



Results:

- 25% increase in skin protection
- 60% of volunteers perceived a protective effect

PEPHA®-PROTECT delays perceptible skin reddening and irritation while at the same time delivering protection to the deeper layers of the skin.

For more information, please visit www.dsm.com/personal-care

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